Output Distillation of Vector-Quantized Diffusion Transformers

Egor Podyablonskiy
Faculty of Computational Mathematics and Cybernetics
Lomonosov Moscow State University
Scientific advisor: Aibek Alanov

Abstract

One hypothesis is tested: codebook-only output distillation improves a vector-quantized diffusion transformer at fixed logical weight storage. Starting from a transferred, layerwise-calibrated VQDM representation of DiT-XL/2, only codebook values are optimized for 500 updates. Assignments, row scales, biases and other model parameters remain fixed. In a matched ImageNet256 evaluation with 50 DDPM steps, guidance 1.5 and 10,000 images per model, FID decreases from 15.40 to 7.81 at 4.15 bits per DiT parameter. The full-precision teacher reaches 6.10. The result supports the hypothesis in this fixed setting, without establishing a new distillation method or robustness across independent runs.

1 Introduction and goal

Diffusion transformers provide a useful generation model but require substantial weight storage [5]. Vector quantization replaces groups of weights with references to small codebooks. Errors left by independent layer calibration may interact through the complete network, motivating an additional optimization stage.

Goal. Reduce the generation-quality loss of a vector-quantized DiT without increasing its logical weight-storage budget.

Main idea. Use the original denoiser as a frozen teacher and jointly adjust the quantized model's codebooks to match its complete output. Keep the representation, assignments and other weights fixed. This is the single idea examined throughout the article and in the accompanying one-slide summary.

2 Related work

DiT uses a transformer denoiser on latent image patches [5]. VQDM combines vector weight quantization with global teacher-output fine-tuning for SDXL and SDXL-Turbo [3]. Thus global distillation is established prior work. Here its restricted transfer to DiT is tested: only codebooks are updated, while the control contains the transferred layerwise stage without subsequent global fine-tuning. This control is not called a complete reproduction of the original VQDM pipeline.

Knowledge distillation transfers a teacher's behavior to a student [4]; the present objective is denoiser-output regression. VQ4DiT instead calibrates codebooks and candidate assignments through transformer-block outputs [1]. It provides relevant context but is not a reproduced baseline in this experiment. Quality measurements use the official ADM toolkit [2]. All conclusions below compare our own matched measurements.

Preprint.

3 Problem statement

Hypothesis. For the same layerwise-quantized initialization, architecture and storage precision, updating only codebook values against full-network teacher predictions lowers generation FID while preserving logical storage.

Model. The teacher is the public class-conditional DiT-XL/2 checkpoint for ImageNet 256×256. The student has the same architecture. Five matrices per block are quantized: attention QKV and output projections, both MLP projections, and adaLN modulation. The 28 blocks contain 140 such matrices and 668,860,416 quantized weights. Activations, biases and parameters outside these matrices are not quantized.

A length-d weight subvector is reconstructed as

$$\widehat{w}_{\ell,r,g} = s_{\ell,r} \sum_{m=1}^{M} C_{\ell,m}[a_{\ell,r,g,m}], \qquad M = 4,\ K = 256,\ d = 8. \tag{1}$$

Here ℓ denotes a layer, r an output row, g a subvector, C codebooks, a stored indices and s row scales. Indices require nominally $M \log_2 K/d = 4$ bits per quantized weight. Including codebooks, scales and unchanged parameters gives 4.150078 logical bits per whole-DiT parameter, excluding the VAE.

Let $f_\theta(z_t, t, y) = (\epsilon_\theta, v_\theta)$ be the teacher and $f_{Q(C)}$ the student. The second head v is the raw learned-variance prediction, not the variance itself. For a real-image VAE latent z_0, inputs are sampled using

$$z_t = \sqrt{\bar{\alpha}_t} z_0 + \sqrt{1 - \bar{\alpha}_t}\, \eta, \qquad \eta \sim \mathcal{N}(0, I), \quad t \sim \mathrm{Uniform}\{0, \ldots, 999\}. \tag{2}$$

Both networks receive the same (z_t, t, y). The train/reserved split is fixed. This KD stage uses real-image latents and is not data-free.

Quality criteria. The primary criterion is lower FID on a matched 10,000-image evaluation, subject to unchanged logical bpp. FID compares means and covariances of generated and reference Inception features. Lower sFID is an auxiliary spatial-feature criterion; higher IS, precision and recall are also reported. Precision and recall describe different aspects of sample fidelity and distribution coverage. A reduction of teacher-output MSE is a separate diagnostic, not a substitute for generation metrics. The experiment tests an exploratory hypothesis in one fixed configuration, not a statistical claim across training seeds.

4 Solution: properties and algorithm

Training objective. Codebooks minimize

$$\mathcal{L}_{\mathrm{KD}} = \mathrm{MSE}(\epsilon_{Q(C)}, \epsilon_\theta) + 0.1\, \mathrm{MSE}(v_{Q(C)}, v_\theta). \tag{3}$$

Teacher parameters, assignments, row scales, biases and other weights remain fixed. The target is the teacher's prediction, rather than the sampled noise η. The variance term is regression on the teacher head, not a DDPM variational-bound loss. Classifier-free guidance is not applied in the training objective; labels are replaced by the null class with probability 0.1.

Property 1: unchanged logical storage. For matrices of size $o_\ell \times i_\ell$, a fixed representation has storage

$$B_{\mathrm{bits}} = \sum_\ell \frac{o_\ell i_\ell}{d} M \lceil \log_2 K \rceil + 16 \sum_\ell MKd + B_{\mathrm{fixed}}. \tag{4}$$

The first term stores indices, the second FP16 codebooks, and B_{fixed} contains unchanged scales and other parameters. None of these array sizes or precisions depends on the updated codebook values. Therefore codebook-only optimization preserves logical bpp. This is a counting argument for the packed representation, not a bound on serialized checkpoints or inference time.

Property 2: a prediction-error bound. Write $S = \epsilon_{Q(C)}$, $T = \epsilon_\theta$ and let D be the number of noise-output elements. On the common input distribution, assume finite second moments and define

$$R(S) = \mathbb{E}\|S - \eta\|^2/D, \qquad D_{\mathrm{KD}} = \mathbb{E}\|S - T\|^2/D. \tag{5}$$

The triangle inequality in L^2 gives

$$\sqrt{R(S)} \le \sqrt{D_{\mathrm{KD}}} + \sqrt{R(T)}. \tag{6}$$

Indeed, $S - \eta = (S - T) + (T - \eta)$. Distillation controls one term in this upper bound; it does not guarantee a reduction of actual noise error or FID. The latter is checked by generation.

Algorithm.

1. Load the completed layerwise VQDM student and its full-precision teacher. Freeze all student parameters except codebook master values.
2. For each of 500 updates, sample a batch of 16 training latents, a time and Gaussian noise, and construct inputs by Equation (2). Apply null-class replacement with probability 0.1.
3. Evaluate teacher and student on identical inputs. Compute Equation (3), backpropagate through the entire student, and update codebooks using Adam with learning rate 10^{-5}.
4. Monitor a fixed held-out teacher-noise probe. Export FP16 codebooks at step 500 and check that assignments and frozen parameters are unchanged. The final500 checkpoint is fixed before the 10,000-image FID calculation.
5. Generate matched image sets for teacher, layerwise-only student and KD student, then evaluate the same external criteria for all three.

FP32 master values are optimized with autograd through FP16 casts; reconstructed weights are used in FP32 computation. The implementation's cast gradients are used, without claiming exact differentiation of a discrete rounding map. Gradients are not unrolled through the complete sampling trajectory.

5 Experiment

Objectives. Test whether the added KD stage lowers generation FID from the same quantized initialization, verify the storage invariant, and examine auxiliary metrics and the remaining gap to the teacher. The full-precision model is a quality reference; the layerwise-only student is the direct control.

Calibration and training data. Initial layerwise calibration used 256 teacher-generated DDPM trajectories. KD uses 5,000 fixed ImageNet training images selected from two fixed shards and encoded by the MSE VAE: 4,000 latents for updates and 1,000 reserved. A fixed probe of 96 input/noise pairs at six timesteps is drawn from the reserved subset. Its MSE is not an average over all 1,000 reserved images. The saved teacher-noise probe is monitored at updates 0, 50, 100, 200 and 500. Final500 also has the lowest saved probe error, without selection by FID.

Evaluation protocol. Each model generates 10,000 images, ten per ImageNet class, using common sample records and paired initial/per-step noise, with selection seed 20261103. Generation uses **DDPM50**, CFG 1.5, `sd-vae-ft-mse`, FP32 computation, disabled TF32 and mathematical attention. The 1,000-step training noising schedule is distinct from these 50 generation steps.

The official ADM Inception graph and ImageNet256 reference are fixed. FID/sFID use stored full-dataset reference statistics; precision/recall use its 10,000 reference images. Each archive extends an earlier same-protocol 5,000-image prefix. All reported metrics are recomputed on the full 10,000-image sets. No result on 5,000 images is compared with a 10,000-image control.

6 Results and error analysis

Table 1: Model–data–criteria summary. IN256/10k denotes ImageNet 256×256, 10,000 generated images per model. All rows use DDPM50, CFG 1.5 and the same MSE VAE. Logical bpp excludes the VAE.

Model	Evaluation data	bpp	FID↓	sFID↓
FP teacher	IN256 / 10k	32	6.098	21.675
Layerwise VQDM	IN256 / 10k	4.1501	15.402	27.601
VQDM + KD500	IN256 / 10k	4.1501	7.814	23.367

Table 2: Auxiliary quality criteria on the same ImageNet256/10k sets as Table 1. Higher values are preferred.

Model	IS↑	Precision↑	Recall↑
FP teacher	253.827	0.7904	0.6659
Layerwise VQDM	145.312	0.6491	0.7115
VQDM + KD500	211.549	0.7427	0.6770

Effect and remaining error. FID decreases from 15.40 to 7.81, a relative reduction of 49.3%, while logical storage remains 4.15 bpp. The saved check reports zero changes among 334,430,208 assignments. sFID, IS and precision improve, but recall falls from 0.7115 to 0.6770. The result therefore includes a coverage trade-off. The KD student's FID remains 1.72 above its own teacher.

The held-out teacher-noise MSE decreases from 0.001162851 to 0.001007803, or 13.33%. This same-input prediction error and the generation FID quantify different effects. Their improvements are reported separately.

Measurement and uncertainty. Independent replay with a freshly fetched official ADM CLI reproduces checked FIDs within 10^{-5}, including a published fixed ADM fixture. Some recall values differ at the fourth decimal place. This validates the checked FID calculation, not bitwise identity of every metric or reproduction of another paper's compression results.

Only one fixed quantizer initialization and optimization setting is considered. No confidence interval across independently trained models is estimated, and a single generated cohort is reused exploratorily. Training data comes from two fixed shards; class-balanced evaluation differs from natural ImageNet frequencies. There is no compute-matched alternative fine-tuning control, so the result demonstrates the benefit of adding this KD stage without establishing its superiority to every other use of 500 updates.

7 Conclusion

The hypothesis is supported in the tested configuration: codebook-only output distillation lowers FID from 15.40 to 7.81 at unchanged logical storage. A residual gap to FP and a recall decrease remain. Distillation itself is established prior work; the completed experiment checks its restricted transfer to DiT. Independent repeats and a compute-matched objective comparison are needed before a broader conclusion.

Artifacts. The manuscript, one-slide summary, three-minute talk and numeric snapshot are available in the practicum repository. Scientific code and experiment history remain in the research repository. This is an intermediate practicum manuscript.

References

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